

HANNA PARK

US Citizen | hpark4@caltech.edu | [linkedin.com/in/park-hanna](https://www.linkedin.com/in/park-hanna) | hannapark1206.github.io

Mechanical engineer interested in robotics, with research experience building and testing aerial and multi-modal robot platforms through flight-control integration, prototyping, and aerodynamic testing.

EDUCATION

Caltech | *B.S. in Mechanical Engineering* | **GPA: 4.3/4.0** | Expected 2027 Pasadena, CA
Relevant Coursework: Robotics, Dynamics & Control, Mechanical Prototyping, Design & Fabrication, CFD, Diff. Equations & Transform Methods
In Progress (2026–27): Machine Learning, Linear Systems & Control, Advanced Fluid Mechanics, Data-Driven Modeling & Dimensional Analysis
University of Cambridge, St John's College | *Engineering* | Sept–Dec 2025 Cambridge, United Kingdom
Completed Courses (graduate-level): Aircraft Aerodynamics & Design; Stability & Control; Nuclear Reactor Engineering; Technology Management

AWARDS

Henry Ford II Scholar Award | *Caltech Engineering and Applied Science, strongest third-year academic record* 2026
Jack E. Froehlich Memorial Award Nominee | *Top 5% GPA, with selection based on academics and research* 2026

SKILLS

Robotics & Flight Control: flight-control integration, actuator control, ROS, MAVLink/MAVROS, ArduPilot/PX4
UAS Tools & Testing: QGroundControl, Mission Planner, OptiTrack, wind-tunnel testing, PIV/dye visualization
Mechanical Design: SolidWorks, engineering drawings, machining, 3D printing, soldering, wiring
Simulation & Programming: OpenFOAM, ANSYS Mechanical, Simulink, Python, MATLAB, C/C++, Linux, Git
Languages: English (native), Korean (native), Japanese (proficient), Spanish (conversational)

EXPERIENCE

SpaceX Engineering Internship Summer 2026
Falcon Propulsion Engineering Intern Hawthorne, CA
M4 (Multi-Modal Mobility Morphobot) – Air–Ground Transforming Robot and Deflector Design January 2026 – Present
California Institute of Technology Pasadena, CA

- Building a custom quadrotor aerial platform for the M4 air-ground robot, including CAD, fabrication, four-motor propulsion/ESC integration, two-servo deflector actuation, avionics wiring, and PX4-based Pixracer Pro setup.
- Designing thrust-biasing deflectors; using ANSYS and OpenFOAM CFD, along with bench testing, to quantify thrust and drag.

Gharib Group Researcher – UAV Flight Testing and Experimental Flow Facilities February 2024 – Present
Center for Autonomous Systems and Technologies – Graduate Aerospace Laboratories of Caltech (GALCIT) Pasadena, CA

- Integrated embedded sensing, flight-control, actuator, and ground-station systems for fixed-wing aerial robot testing.
- Built ROS/MAVLink workflows for synchronized pressure, IMU, and autopilot data during controlled free-flight experiments.
- Designed and wind-tunnel calibrated five-hole Pitot probe hardware for real-time aerodynamic flow measurement.
- Built water-tunnel hardware, turbine arrays, and Arduino/PWM control for PIV/dye studies of shear-layer and vortex shedding.

Wellatex INC. – Assembly Machine Management and Product Design Lead January 2019 – Present

- Assembled industrial packing machinery and designed palletized shipping boxes for 50+ product dimensions.

Golwala Group Summer Undergraduate Research Fellowship – Cryogenic Detectors June 2024 – September 2024
California Institute of Technology Pasadena, CA

- Redesigned cryogenic detector hardware through CAD, circuit-geometry updates, mounting design, and wire-bonded assembly.
- Developed a Python resonance-fitting pipeline for superconducting resonator analysis, reducing fit time by 80%.

LEADERSHIP, CLUBS, AND INTERESTS

Teaching Assistant, Biomechanics (Prof. Michael Dickinson) | *Caltech* January 2025 – March 2025

- Prepared and graded exams; held office hours to support student understanding of solid/fluid mechanics and materials.

Caltech Air and Outer Space Club | *NASA Big Idea Challenge Finalist – Awarded \$150,000* Fall 2023 – Winter 2024

- Developed proposal for an inflatable lunar landing shield, modeling dust/regolith behavior during landing.

Caltech Admissions Ambassador Professional Development Lead May 2024 – Present

- Conduct tours and serve as a student panelist; deliver audience-specific Q&A on curriculum, research, and student life.

Interests | *Cello and baking*

- Play cello in chamber/orchestra settings and enjoy baking.